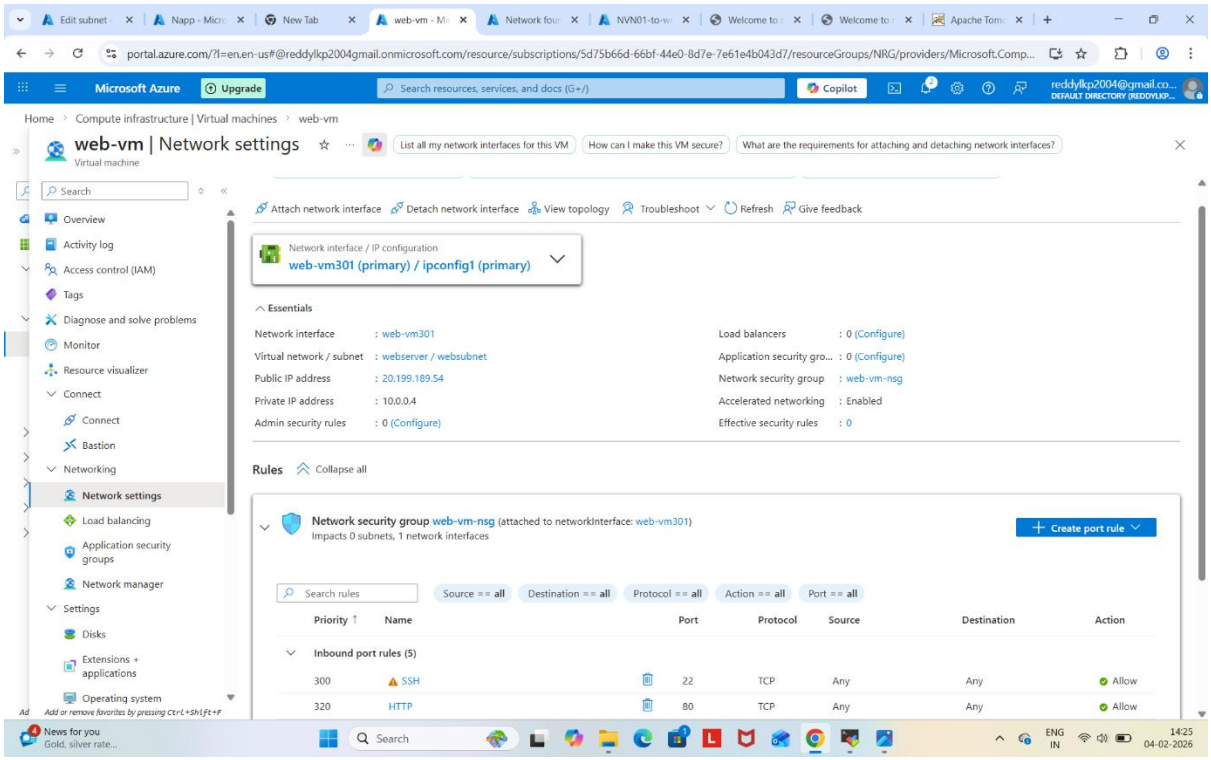
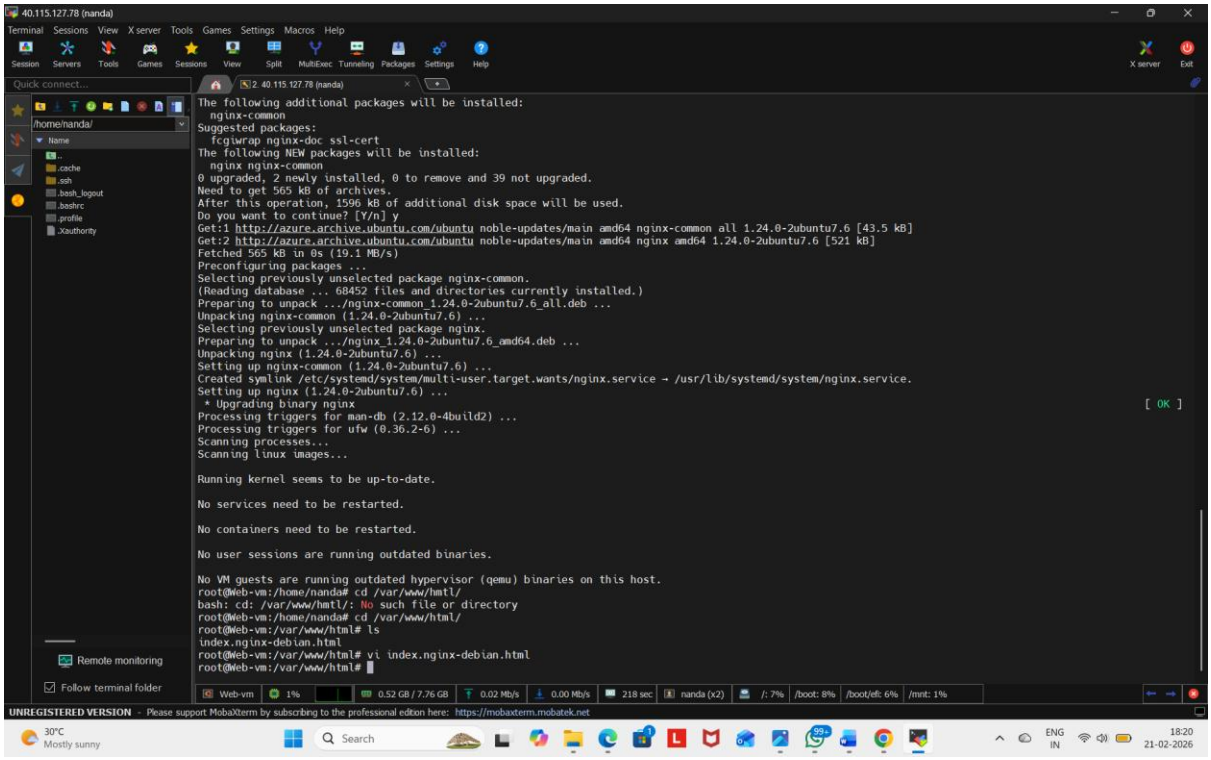
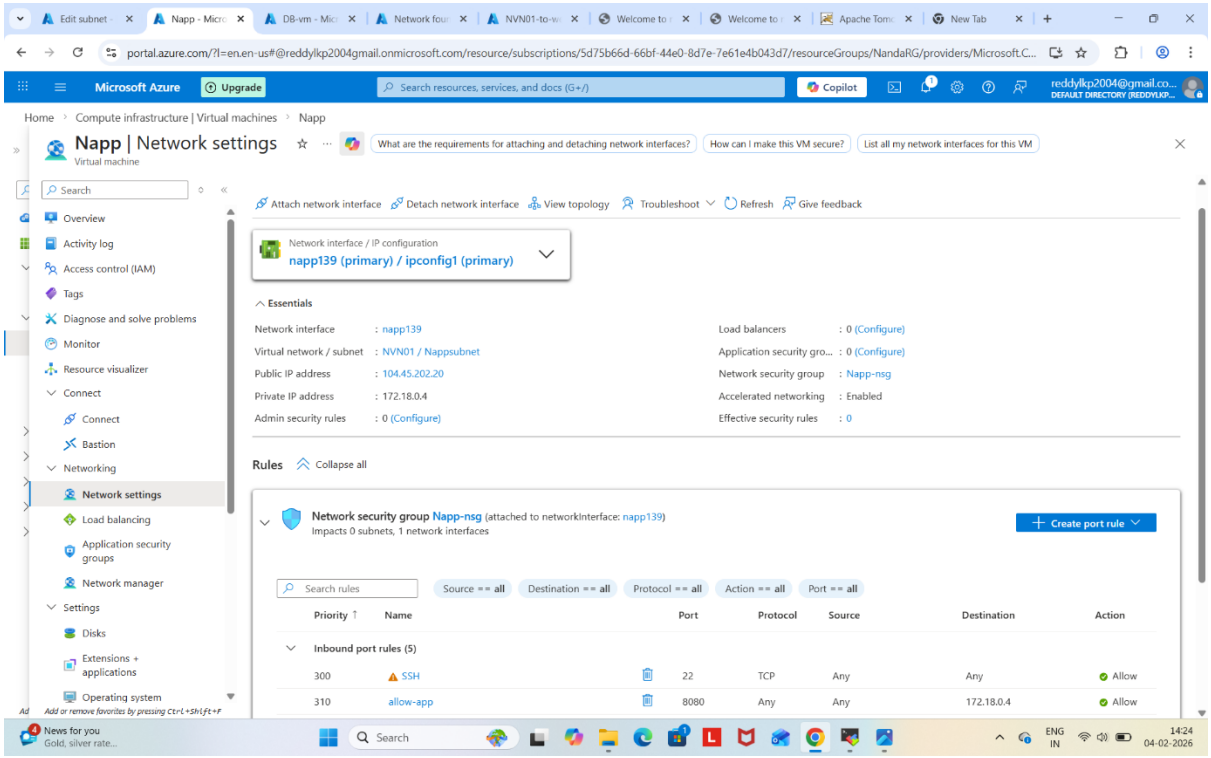
3 TIER ARCHITECTURE (WEB-APP-DB)

- Create a **Web01** and installed (**Nginx**) into it and **Nsg** (source: Any, Dest: App01,Dport:80) & (SSH:22)
- Allow port from source: **Web01**(private Ip),Dest: **Napp**(private Ip), Dport:**8080**





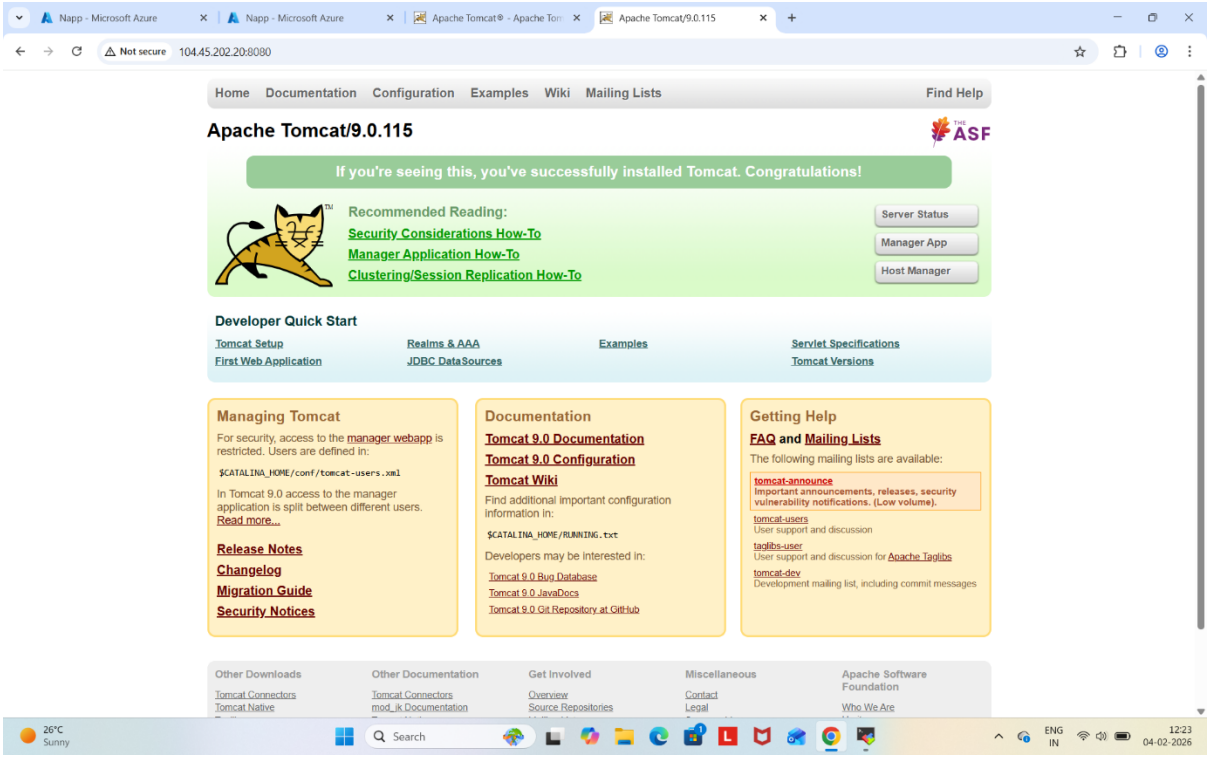
➢ Next App01 installed (Tomcat)



```
the root password from the network.

Disallow root login remotely? (Press y|Y for Yes, any other key for No) : y
Success.

By default, MySQL comes with a database named 'test' that
anyone can access. This is also intended only for testing,
and should be removed before moving into a production
environment.


Remove test database and access to it? (Press y|Y for Yes, any other key for No) : y
 - Dropping test database...
Success.

 - Removing privileges on test database...
Success.

Reloading the privilege tables will ensure that all changes
made so far will take effect immediately.

Reload privilege tables now? (Press y|Y for Yes, any other key for No) : y
Success.

All done!
root@DB-vm:/home/nanda# cd /etc/mysql/mysql.conf.d/
root@DB-vm:/etc/mysql/mysql.conf.d# ls
mysql.cnf  mysqld.cnf
root@DB-vm:/etc/mysql/mysql.conf.d# vi mysqld.cnf
\root@DB-vm:/etc/mysql/mysql.conf.d# systemctl restart mysql
root@DB-vm:/etc/mysql/mysql.conf.d# systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-04 07:06:35 UTC; 21s ago
    Process: 5669 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
   Main PID: 5678 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 9507)
     Memory: 363.1M (peak: 380.7M)
        CPU: 585ms
     CGroup: /system.slice/mysql.service
             └─5678 /usr/sbin/mysqld

Feb 04 07:06:34 DB-vm systemd[1]: Starting mysql.service - MySQL Community Server...
Feb 04 07:06:35 DB-vm systemd[1]: Started mysql.service - MySQL Community Server.
root@DB-vm:/etc/mysql/mysql.conf.d#
```

```
apache-tomcat-9.0.115/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
apache-tomcat-9.0.115/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
apache-tomcat-9.0.115/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
apache-tomcat-9.0.115/webapps/manager/WEB-INF/jsp/sessionsList.jsp
apache-tomcat-9.0.115/webapps/manager/WEB-INF/web.xml
apache-tomcat-9.0.115/webapps/manager/css/manager.css
apache-tomcat-9.0.115/webapps/manager/images/asf-logo.svg
apache-tomcat-9.0.115/webapps/manager/images/favicon.ico
apache-tomcat-9.0.115/webapps/manager/images/tomcat.svg
apache-tomcat-9.0.115/webapps/manager/index.jsp
apache-tomcat-9.0.115/webapps/manager/status.xsd
apache-tomcat-9.0.115/webapps/manager/xform.xsl
apache-tomcat-9.0.115/bin/catalina.sh
apache-tomcat-9.0.115/bin/ciphers.sh
apache-tomcat-9.0.115/bin/configtest.sh
apache-tomcat-9.0.115/bin/daemon.sh
apache-tomcat-9.0.115/bin/digest.sh
apache-tomcat-9.0.115/bin/makebase.sh
apache-tomcat-9.0.115/bin/setclasspath.sh
apache-tomcat-9.0.115/bin/shutdown.sh
apache-tomcat-9.0.115/bin/startup.sh
apache-tomcat-9.0.115/bin/tool-wrapper.sh
apache-tomcat-9.0.115/bin/version.sh
root@Napp:/home/nanda# ls
apache-tomcat-9.0.115  apache-tomcat-9.0.115.tar.gz
root@Napp:/home/nanda# mv apache-tomcat-9.0.115 tomcat
root@Napp:/home/nanda# ls
apache-tomcat-9.0.115.tar.gz  tomcat
root@Napp:/home/nanda# cd tomcat/
root@Napp:/home/nanda/tomcat# ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
root@Napp:/home/nanda/tomcat# cd bin/
root@Napp:/home/nanda/tomcat/bin# ls
bootstrap.jar       ciphers.bat                 configtest.bat  digest.sh        setclasspath.sh  startup.sh            tool-wrapper.sh
catalina-tasks.xml  ciphers.sh                  configtest.sh   makebase.bat     shutdown.bat     tomcat-juli.jar       version.bat
catalina.bat        commons-daemon-native.tar.gz daemon.sh      makebase.sh      shutdown.sh      tomcat-native.tar.gz  version.sh
catalina.sh         commons-daemon.jar          digest.bat      setclasspath.bat startup.bat      tool-wrapper.bat
root@Napp:/home/nanda/tomcat/bin# ./startup.sh
Using CATALINA_BASE:   /home/nanda/tomcat
Using CATALINA_HOME:   /home/nanda/tomcat
Using CATALINA_TMPDIR: /home/nanda/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /home/nanda/tomcat/bin/bootstrap.jar:/home/nanda/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@Napp:/home/nanda/tomcat/bin#
```

➢ Tomcat Installed



TOMCAT

To install **Apache Tomcat** on **Ubuntu**, follow these steps:
To switch from a **normal user** ($ prompt) to the **root user** (# prompt) in Ubuntu, use any of the following commands:

Method 1: Using sudo su (Recommended)

- sudo su
  a. This switches you to the **root user**.
  b. The prompt will change from $ to #.

Step 1: Update System Packages

- apt update

Step 2: Install Java

Tomcat requires Java. Install OpenJDK:

- apt install default-jdk -y

Verify the installation:

- java -version

Step 3: Download Tomcat

- Go to the Apache Tomcat official website and get the latest version.

Run:

- wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.18/bin/apache-tomcat-10.1.18.tar.gz

(Replace the version number with the latest one if needed.)

- ls (to list the files)

Step 4: Extract and Configure Tomcat

- tar -xvzf apache-tomcat-10.1.18.tar.gz
- ls
- mv apache-tomcat-10.1.18 tomcat
- ls

To start Apache Tomcat manually from the **bin** directory, follow these steps:

Step 1: Navigate to the Tomcat Directory

- cd /tomcat/bin

Step 2: Start Tomcat

Run the startup script: (./ is to execute a command)

- ./startup.sh

Step 4: Access Tomcat

Open your browser and go to:
http://your_server_ip:8080

➢ Editing context.xml for Host Manager:

The Valve element in Tomcat's context.xml file is used for **access control and request filtering**. Specifically, the RemoteAddrValve restricts access based on the client's IP address.

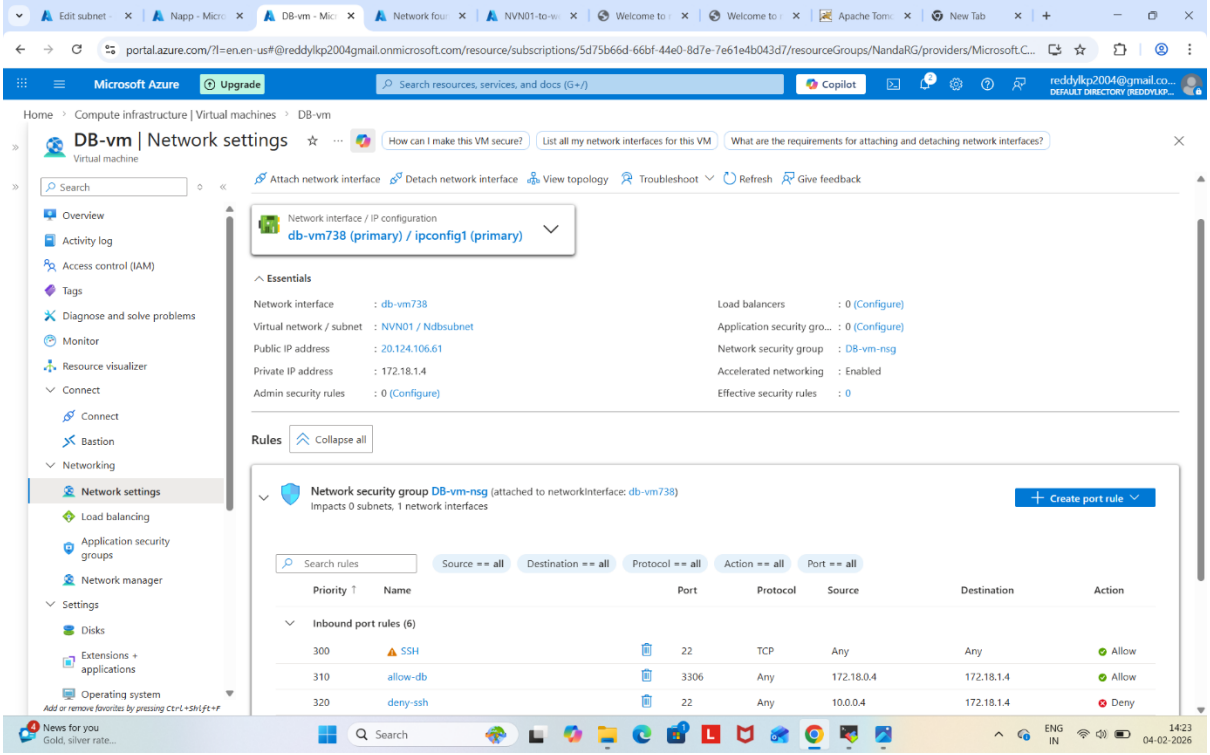
Use of Valve in context.xml (META-INF)

Purpose:

- Controls access to web applications (like manager and host-manager).
- Allows or denies requests from specific IP addresses.
- Enhances security by limiting administrative access.

- Next **DB01** Install (**Mysqldb**)
- Allow port from source: **App01**(private Ip),

  Dest: **DB01**(private Ip),Dport: **3306**



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MYSQL

To install MYSQL on **Ubuntu**, follow these steps:
To switch from a **normal user** ($ prompt) to the **root user** (# prompt) in Ubuntu, use any of the following commands:

Method 1: Using sudo su (Recommended)

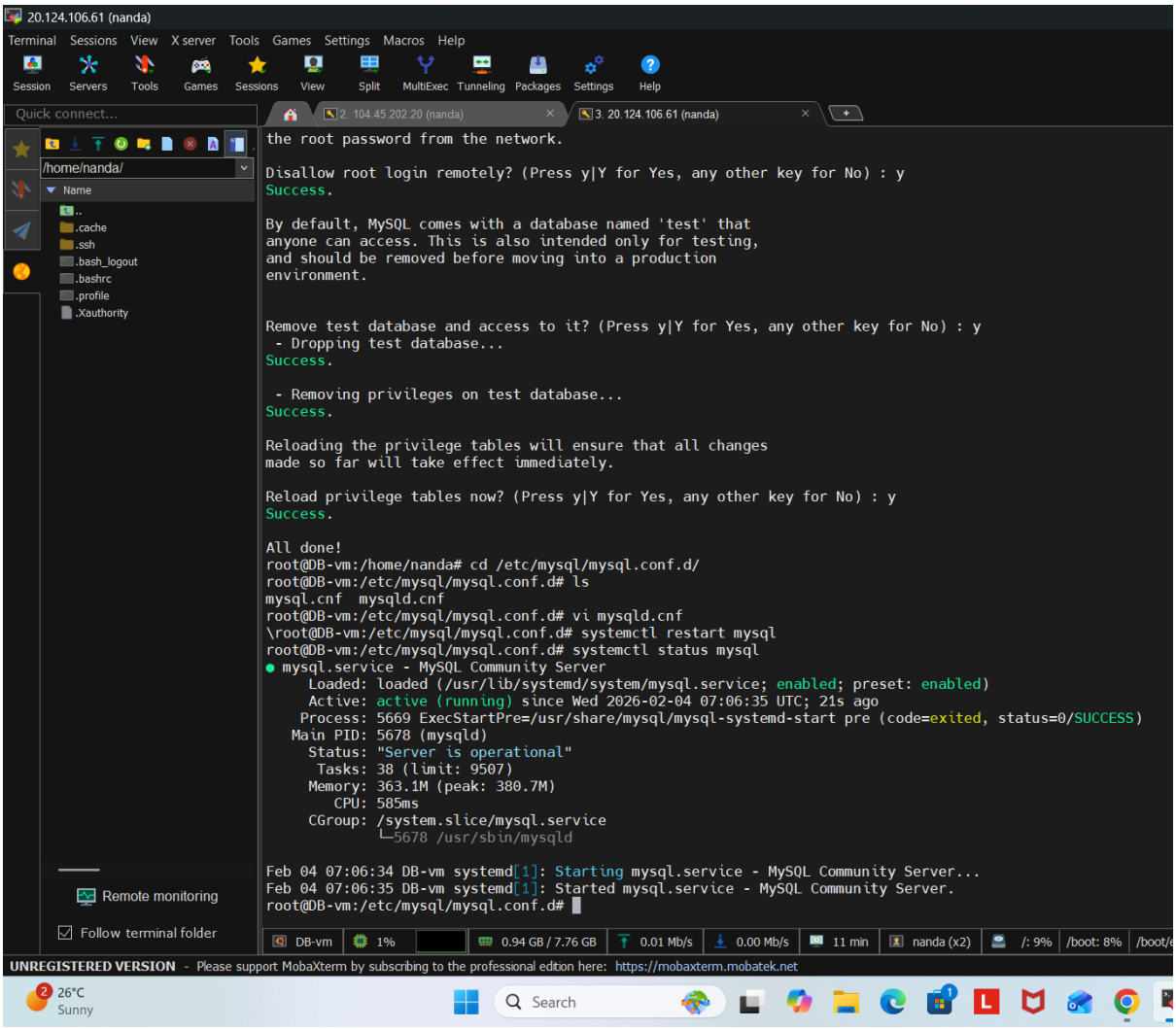
- sudo su
  - c. This switches you to the **root user**.
  - d. The prompt will change from $ to #.

Step 1: Update System Packages

- apt update
- Run the following command to install MySQL:
- apt install mysql-server –y
- To improve security, run the following command:
- mysql_secure_installation
- Edit the MySQL configuration file using a text editor:
- nano /etc/mysql/mysql.conf.d/mysqld.cnf
- Locate the following line: bind-address = 127.0.0.1
- bind-address = 0.0.0.0
- EX: Modify it based on your requirement:

Scenario	Configuration
Listen on all interfaces (for remote access)	bind-address = 0.0.0.0
Listen on a specific IP (e.g., 192.168.1.100)	bind-address = 192.168.1.100

- Save and exit (CTRL + X, then Y, then Enter).
- Apply the changes by restarting MySQL:
- systemctl restart mysql
- Check if MySQL is running:
- systemctl status mysql

After Installing mysql on DB, Edit the file which is stored inside /etc/mysql/mysql.conf.d/mysql.cnf This is the file we have to edit, In side there is a **Bind-address** it should be 0.0.0.0, save & exit.

- Now see **connection** from **App01** to **DB01** with the help of Telnet DB (private Ip)(3306) from App01, Because Db we can't browse on internet, That's y we coonect from App to DB.
- Next configure **Deny** access from **web01** to **DB01**
- No-direct Connection b/w them